

Elements of Statistics and Probability

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Conditional Probability

- Conditional probability is defined as the likelihood of an event or outcome occurring, based on the occurrence of a previous event or outcome. The probability that event A occurs, given that event B has occurred, is called a conditional probability
- The conditional probability of A, given B, is denoted by the symbol $P(A|B)$, is given by

$$P(A|B) = \frac{P(A \cap B)}{P(B)}; \text{ for } P(B) > 0$$

Product Rule or Law of Multiplication

- For two **dependent** events A and B:
 - $P(A \cap B) = P(A|B)P(B)$
 - $P(A \cap B) = P(B|A)P(A)$
- For two **independent** events A and B:
 - $P(A \cap B) = P(A)P(B)$

Example

- In a class of 120 students, 60 are studying English, 50 are studying French and 20 are studying both English and French. If a student is selected at random from this class, what is the probability that he or she is studying English given that he is studying French

Solution

- Here, $P(E) = \frac{60}{120} = 0.5$
- $P(F) = \frac{50}{120} = 0.42$
- $P(E \cap F) = \frac{20}{120} = 0.17$
- The probability that s/he is studying English given that s/he is studying French is,

$$P(E|F) = \frac{P(E \cap F)}{P(F)} = \frac{20}{50} = 0.4$$

Chain Rule

- From the definition of conditional probability

$$P(A \cap B) = P(A|B)P(B)$$

- Similarly, for events A, B & C

$$P(A \cap B \cap C) = P(A)P(B|A)P(C|A \cap B)$$

Example

- In rainy season, it rains 70% of the days in Bangladesh. When it rains, 80% times it makes thunderstorms. What is the probability that, in a particular day of rainy season, it will rain and it will thunderstorm?

Solution

- Let, R = it will rain on that particular day
- T = it will thunderstorm on that particular day
- Here, $P(R) = \frac{70}{100} = 0.7$ and $P(T|R) = \frac{80}{100} = 0.8$
- Therefore, the probability that, on that particular day of rainy season, it will rain and it will thunderstorm is

$$P(R \cap T) = P(T|R)P(R) = 0.8 \times 0.7 = 0.56$$

Probability Using Contingency Tables

- A general 2×2 contingency table:

	Y_1	Y_2	Total
X_1	a	b	$a + b$
X_2	c	d	$c + d$
Total	$a + c$	$b + d$	$a + b + c + d$

- Here, X and Y are two variables, and each of them has two possible categories

Example

- Consider the contingency table for smoking and cancer status:

	Cancer	Not cancer	Total
Smoker	7860	1530	9390
Non-smoker	5390	11580	16970
Total	13250	13110	26360

- (a) Probability that a randomly selected person is a smoker?
- (b) Probability that a randomly selected person has cancer?
- (c) Probability that a randomly selected person is both a smoker and a cancer patient?
- (d) If a person is a smoker, what is the probability that he is a cancer patient?

Tree Diagram for Conditional Probability

- To make a probability tree diagram, you need to find all the possible outcomes and their associated probabilities. Follow these steps:
 - 1 Decide if the events depend on each other or happen independently
 - 2 Draw branches to show the initial outcomes
 - 3 Put the probabilities for each outcome on its branch
 - 4 Draw additional branches for the next outcomes, considering whether events are dependent or independent. Include the associated probabilities¹

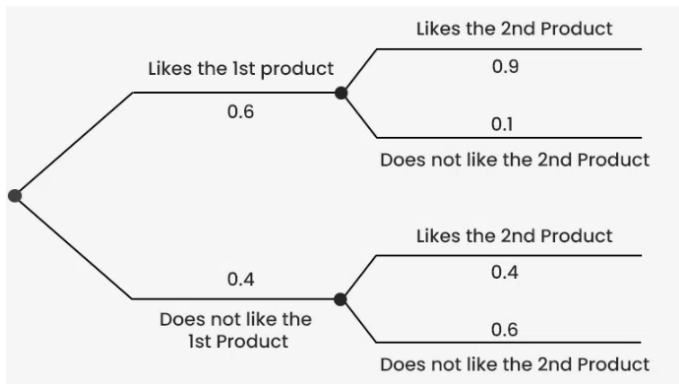
Tree Diagram for Conditional Probability (cont.)

- 4 Repeat step 4 as needed
- 5 To find the total probability, multiply the probabilities along the branches. The sum of these probabilities should always be 1

Example

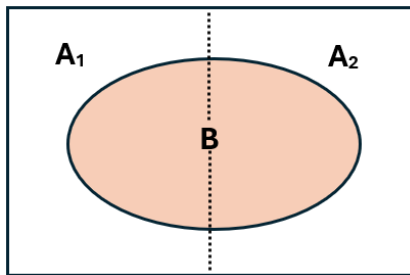
- Suppose a person buying two different products online. The probability of them liking the first product is 0.6. If they like the first product, the chance they'll also like the second one is 0.9. However, if they don't like the first product, the probability of liking the second one drops to 0.4. To find the conditional probability of disliking both products, we can use a probability tree diagram:

Example (cont.)



Law of Total Probability

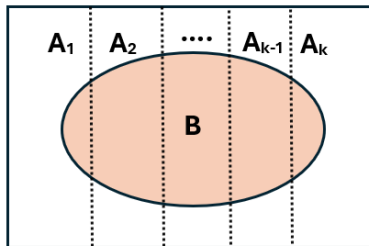
- Let, events A_1 and A_2 form partition of S . Let B be an event with $P(B) > 0$



- Then
$$P(B) = P(A_1 \cap B) + P(A_2 \cap B)$$
$$= P(A_1)P(B|A_1) + P(A_2)P(B|A_2)$$

Law of Total Probability

- Let, events $A_1, A_2, \dots, A_k$ form partition of S . Let B be an event with $P(B) > 0$



- Then

$$\begin{aligned} P(B) &= P(A_1 \cap B) + P(A_2 \cap B) + \dots + P(A_k \cap B) \\ &= P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + \dots + P(A_k)P(B|A_k) \\ &= \sum_{i=1}^k P(A_i)P(B|A_i) \end{aligned}$$

Bayes' Theorem

- If A and B are two events, then

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

- Proof:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$$

Bayes' Theorem

- If there are three events A_1 , A_2 , and B where A_1 and A_2 are mutually exclusive and collectively exhaustive events then the workable form of Bayes' theorem is,

$$\begin{aligned}P(A_1|B) &= \frac{P(B|A_1)P(A_1)}{P(B)} \\&= \frac{P(B|A_1)P(A_1)}{P(B|A_1)P(A_1) + P(B|A_2)P(A_2)}\end{aligned}$$

Bayes' Theorem

- Thus, for n events $A_1, A_2, \dots, A_m, \dots, A_n$, the form of Bayes' theorem is,

$$\begin{aligned} P(A_m|B) &= \frac{P(B|A_m)P(A_m)}{P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + \dots + P(B|A_n)P(A_n)} \\ &= \frac{P(B|A_m)P(A_m)}{\sum_{i=1}^n P(B|A_i)P(A_i)} \end{aligned}$$

Example 1

- A consulting firm submitted a bid for a large consulting contract. The firm's management felt it had a 50–50 chance of landing the project. However, the agency to which the bid was submitted subsequently asked for additional information. Past experiences indicate that for 75% of successful bids and 40% of unsuccessful bids the agency asked for additional information.
 - (a) What is the prior probability of the bid being successful (that is, prior to the request for additional information)?
 - (b) What is the conditional probability of a request for additional information given that the bid will ultimately be successful?
 - (c) Compute the posterior probability that the bid will be successful given a request for additional information.

Solution (Example 1)

- Let,
 - A_1 = the bid is successful
 - A_2 = the bid is unsuccessful
 - B = additional information requested
- Here,
 - $P(A_1) = 0.5$
 - $P(A_2) = 0.5$
 - $P(B|A_1) = 0.75$
 - $P(B|A_2) = 0.4$

Solution (Example 1)

(a) $P(A_1 = 0.5) = 0.5$

(b) $P(B|A_1) = 0.75$

(c)

$$\begin{aligned}P(A_1|B) &= \frac{P(A_1 \cap B)}{P(B)} \\&= \frac{P(B|A_1)P(A_1)}{P(B|A_1)P(A_1) + P(B|A_2)P(A_2)} \\&= \frac{0.75 \times 0.5}{(0.75 \times 0.5) + (0.40 \times 0.5)} \\&= 0.652\end{aligned}$$

Exercise 1

- A particular study showed that 12% of men will likely develop prostate cancer at some point in their lives. A man with prostate cancer has 95% chance of a positive test result from a medical screening exam. A man without prostate cancer has a 6% chance of getting a (false) positive test result. What is the probability that a man has cancer given he has a positive test result?

Example 2

- Suppose that an insurance company classifies people into one of the three classes— good risks, average risks and bad risks. Their record indicate that the probabilities that good, average, and bad risk persons will be involved in an accident over a 1-year span are 0.05, 0.15 and 0.3 respectively 20% of the population are “good risks”, 50% are “average risks”, and 30% are “bad risks”.
 - (a) What proportion of people have accidents in a fixed year?
 - (b) If policy holder A had no accidents in 1987, what is the probability that he or she is a good risk person?

Solution (Example 2)

- Let,
 - A_1 = good risk
 - B_1 = average risk
 - A_3 = bad risk
 - B = accident occurs
- Here,
 - $P(A_1) = 0.2, P(A_2) = 0.5, P(A_3) = 0.3$
 - $P(B|A_1) = 0.05, P(B|A_2) = 0.15, P(B|A_3) = 0.3$

Solution (Example 2)

(a)

$$\begin{aligned}P(B) &= \sum_{i=1}^n P(B|A_i)P(A_i) \\&= P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + P(B|A_3)P(A_3) \\&= (0.05 \times 0.2) + (0.15 \times 0.5) + (0.3 \times 0.3) \\&= 0.175\end{aligned}$$

Solution (Example 2)

- (b)
- Let B' = accident does not occur
 - $P(B') = 1 - 0.175 = 0.825$ and $P(B'|A_1) = 1 - 0.005 = 0.95$

$$\begin{aligned}P(A_1|B') &= \frac{P(B'|A_1)P(A_1)}{P(B')} \\&= \frac{0.95 \times 0.20}{0.825} \\&= 0.23\end{aligned}$$

- If policy holder A had no accidents in 1987, the probability that he or she is a good risk is 23%

Exercise 2

- A diagnostic test has a probability 0.95 of giving a positive result when applied to a person suffering from a certain disease, and a probability 0.10 of giving a (false) positive when applied to a non-sufferer. It is estimated that 0.5% of the population are sufferers. Suppose that the test is now administered to a person about whom we have no relevant information relating to the disease (apart from the fact that he/she comes from this population). Calculate the following probabilities:
 - (a) that the test result will be positive;
 - (b) that given a positive result, the person is a sufferer;
 - (c) that given a negative result, the person is a non-sufferer;
 - (d) that the person will be misclassified.

Thank You